Data Driven Probability of Failure Modelling for Rail Track Assets

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Abstract

Effective railway operators need reliable and explainable risk estimates for asset failure to manage maintenance efficiently. This study presents a data-driven framework for identifying the probability of failure for rail-track segments, using routinely collected inspection data and converting these probabilities to numeric Failure Mode and Effects Analysis (FMEA) Occurrence estimates. The two condition datasets were RailWear (head-loss (%, cross-section loss and gauge/field-face wear) and Geometric Defects (radius, Cant, gauge, alignment). After the data was pre-processed and merged on spatial identifiers, a binary failure label (i.e., Failure) was created based on engineering thresholds (i.e., head loss >15% or vertical wear > 3 mm). A Logistic Regression baseline model and an XGBoost model were trained on the stratified 70/30 split. Both the Logistic Regression and XGBoost were able to produce calibrated probabilities, with strong performance (ROC-AUC ≈ 0.98) across both models, and feature analyses identified head-loss, cross-section loss, gauge, radius and Cant as driving predictors, which was consistent with the expected wear - geometry logic. Predicted probabilities were facilitated and mapped to the FMEA Occurrence scale (i.e., 1-10) to support an objective, data-driven risk ranking of the track segments, and allow integration back into existing reliability frameworks. The contribution in this paper is a transparent, reproducible pipeline that connects interpretable machine learning with FMEA, provides the capability of proactive maintenance planning and creates the foundation for future work that integrates temporal and environmental data and maintenance action data.

Keywords

railway infrastructure, failure probability, machine learning, logistic regression, XGBoost, FMEA Occurrence.

1. Introduction

Railway systems play an essential role for transport and the economy; however, rails, sleepers and ballast are subjected to cyclic loads and environmental impacts, leading to damage, misalignment, and instability, contributing to derailments and delays. Reliability is an ongoing engineering challenge. Maintenance is often based on periodic inspections to identify risks when inspections can provide a level of quantitative data, there is still mostly only qualitative interpretation. Risk ranking using a Failure Mode and Effect Analysis (FMEA) that assesses Severity (S), Occurrence (O) and Detection (D) can assess high risk failure modes. Occurrence ratings are subject to judgment or past counts, which may be inconsistent and limit a projection.

Machine learning (ML) can convert such data into quantitative indicators of reliability. Algorithms that have been trained on historical inspection data will provide an estimation of the likelihood of an asset's failure given the physical observations recorded. Recent studies (e.g., Dai, 2025; Wang, 2022; Wongkaew, 2024) suggests that Logistic Regression and XGBoost are capable of accurately predicting rail defects, yet rarely link the results with FMEA, and the results are left unused for maintenance planning purposes.

The goal of this paper is to fill that gap by developing an integrated FMEA framework connecting ML defect probability measures to the FMEA Occurrence metric to objectively and repeatably establish risk scores. FMEAs may contribute to the field of interpretability but may lack quantitative rigour, and ML may potentially contribute to the field of precision without being counted as practice within engineering. An interpretive framework using FMEA, and a quantitative framework using ML, could potentially merge into a ranking and scheduling of maintenance by rail agencies.

The paper presents a probability-of-failure model which is interpretable and based on data and given values on the FMEA Occurrence scale. The research uses two routinely collected datasets: RailWear (rail-head loss, cross-sectional wear and surface parameters) and Geometric Defects (curvature, Cant, gauge and alignment). Logistic Regression serves as an interpretable baseline and XGBoost is a non-linear comparison. Predicted probabilities are deducted to an FMEA Occurrence scale (1 = rare, 10 = frequent), and subjective ratings are turned into an objective property of risk.

The research questions are:
RQ1. What wear and geometry variables are the strongest predictors of rail failure and are they consistent across Logistic Regression and XGBoost?
RQ2. To what extent can ML models trained on inspection data estimate failure probability as well as methods used in practice which rely on inspection data?
This paper provides a novel contribution to the evidence that routine inspection data can be an objective and repeatable process for FMEA Ocurrence's scores and data-driven maintenance prioritisation. Section 2 reviews related work; a description of the methodology is explained within section 3; results and discussed outcomes are presented in section 4; and implications and future work are outlined in section 5.

2.Related Work

2.1 Introduction

Railway track infrastructure including rails, sleepers, ballast, and geometry are essential for a safe and efficient transport system but are subjected to constant loads, environmental stress, and material deterioration. These assets degrade overtime and could have significant risk of costly damage or derailment unless detected early. Traditional maintenance depends on scheduled inspections, along with qualitative risk scoring based on expert opinion. While these are often a baseline for safety, they are not founded on probabilities and do not account for the active state of asset condition. With the growing availability of data and machine learning (ML) capabilities, we now have an opportunity to effectively quantify the probability of asset failure through models. This section reviews how research is shifting from conventional reliability and FMEA approach towards probabilistic, data-driven prediction, and how those can be integrated.

2.2 Traditional Reliability and FMEA Approaches

Traditional railway maintenance methodologies apply deterministic reliability models and FMEA to assess and prioritise risk. In FMEA, risk is a function of three failure mode ratings: Severity (S), Occurrence (O), and Detection (D) combined as a Risk Priority Number (RPN) that creates ratings established from subjectivity, producing counts that are highly variable and often operate in a reactive manner. FMEA methods, including fuzzy-logic FMEA and sensitivity analyses FMEA, have been developed for use in other studies to create greater consistency in scores, yet the method has not regained any significant traction in railways. Thus, FMEA, to this point, has supplied systematic artifacts documenting the rationale, yet is not predictive, motivating attempts to replace subjective attribution for occurrence with objective, probability rates derived from data.

2.3 Data-Driven Prediction of Rail Failures

The digitisation of inspection processes has prompted greater use of machine-learning algorithms to predict asset deterioration. Dai (2025) tested Random Forest, XGBoost, and Logistic Regression methods on the multi-asset rail datasets and found a relatively high predictive accuracy. However, the research did not integrate the results in a reliability-based framework like FMEA. Wang (2022) examined freight-network rail breaks with Gradient Boosting and exhibited early-warning capability with predictions of failures up to one month where possible. Wongkaew (2024) carried out probabilistic modelling for buckling failure using regression techniques that included environmental variables and noted observed temperature and track-geometry effects, however, the study only considered buckling in a single mode of behaviour. García-Méndez (2025) considered incorporating SHAP explainability to predictive maintenance as a way to create interpretability for the end-user, as well as wider acceptance among stakeholders. Overall, these studies demonstrate predictive accuracy with ML algorithms, but outputs often remain separated from decision-making processes. Predicted probabilities at best can only support prioritised maintenance planning when output is formally integrated within a structured risk framework.

2.4 Integration of Wear and Geometry Parameters

Track deterioration is a function of both mechanical wear and geometric misalignment. Earlier work examined wear and geometry separately, but later studies (Dai 2025; Zhang & Chen 2024) showed that curvature and Cant accelerate deterioration through added stress, confirming that wear and geometry jointly drive rail degradation. This is problematic because public datasets do not reliably contain both of these data-forms to represent accurate measurement due to lack of high-resolutions in both forms, and few studies and predictions have demonstrated the combination of field-proven expected wear and failures, risks that are unified. This research provides a solution for this - to combine the existing RailWear datasets and the Geometric Defects datasets to provide a method of addressing each of the wear and alignment variables within one modelling framework.

2.5 Linking Predictive Models to FMEA and Risk Ranking

Although machine learning models can estimate failure probabilities with high accuracy, their integration into reliability engineering remains limited. Zhao, J. (2023) demonstrated the value of probabilistic outputs derived from an ensemble derailment-risk model, which were then converted to quantitative outputs to demonstrate the potential use of predicting risk probabilities to inform maintenance decision-making. In the railway-reliability literature, Szkoda and Satora (2019) used FMEA to analye rolling-stock systems and Stenstrom (2021) and García-Méndez (2025), where the

authors discussed data-driven improvements to dependability and theoretical improvements to the interpretability of predictive maintenance. Enhanced methodologies for assessing risk associated with FMEA perform better scoring a range of other industries, like fuzzy approaches or the application of a sensitivity analysis (Kumar R., 2018) but are rarely applied in the domain of rail infrastructure. Bayesian reliability models (Zhao, 2023) can improve rail risk assessments FMEA Occurrences scores that would connect FMEA Occurrence scores to Bayesian probability methodology for scoring risk; Bayesian methods also would improve the assessment of subjective bias while scoring the FMEA Occurrence risks. Furthermore, Dai J (2025) suggests mapping probabilities from machine learning directly to the continuous update of the Risk-Priority-Number (RPN). The literature lacks empirical evidence linking ML predicted probabilities and FMEA scores, which this paper provides, allowing for interpretable, data-driven risk ranking for railway maintenance.

2.6 Summary of the Review

Additionally, literatures demonstrate advancements from qualitatively based inspections to data-driven predictive maintenance, however, there exist limitations. Traditional FMEA gives structure but is not quantitatively based, while ML models can be accurate, they struggle with interpretability or operational aspects. There are limited studies that utilise geometry and wear data in a single dataset, and fewer still that map probabilistic predictions to FMEA-based occurrence scores. This study addresses these gaps by using interpretable logistic regression and XGBoost models trained on merged geometry and wear data to predict failure probability and map these outcomes to FMEA-based occurrence scores for operationalised maintenance priority setting. This contribution merges predictive analytics with engineering risk frameworks and enables clear and rational, evidence-based, prioritisation for maintenance on railway track.

3. Methods

3.1 Overview of Research Approach

This research takes a data-driven approach with a component of artefact development. The study is quantitative, using actual inspection data to estimate the chance of rail track failure and to implement a practical artefact, a predictive-risk pipeline that aligns machine-learning (ML) probabilities with FMEA Occurrence scores. The focus of the methodology is on data preparation, modelling and validation instead of experimental manipulation. The research acquires new analytical knowledge about deterioration mechanisms through these processes while simultaneously creating a reusable artefact that facilitates objective, probability-based maintenance planning.

3.2 Research Design and Development

The layout of this research was planned in four consecutive phases of data preparation, feature selection, modelling, and analysis and visualisation as shown in Figure 1. Each phase signifies a unique but connected step taking from raw inspection data to an understandable failure-probability framework fitting for risk evaluation in an FMEA context.

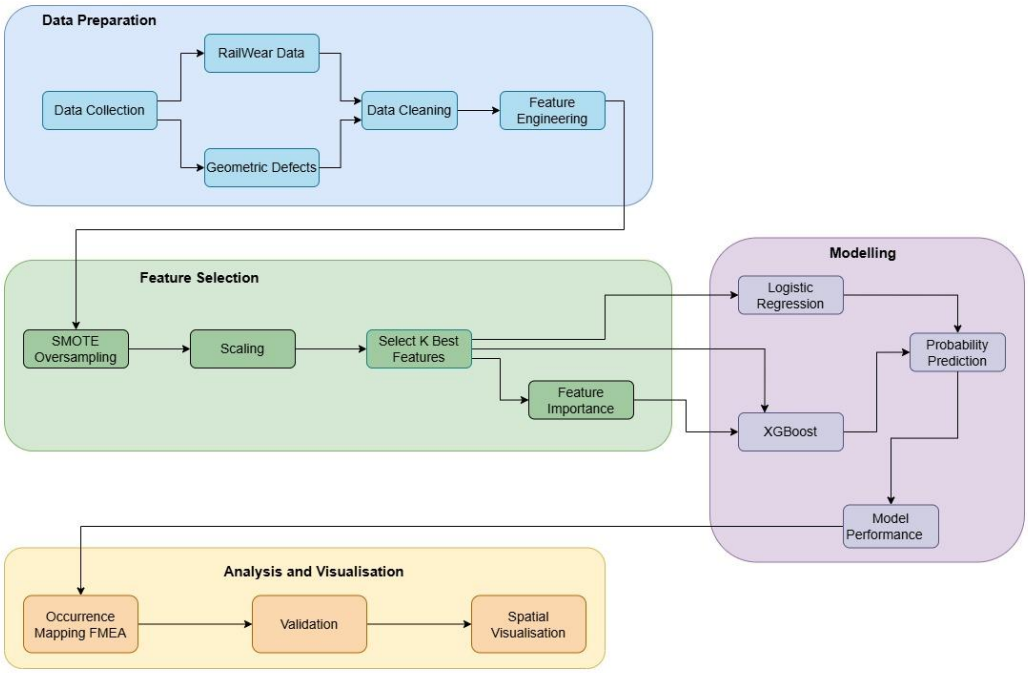

Figure 1. Workflow of the data-driven failure probability modelling process

At first, the datasets RailWear and Geometric Defects were brought in and combined based on common spatial identifiers (TrackCode and KmPost) to make sure that the wear and geometry variables referred to the same segments of the rail. The second stage dealt with class balancing and statistical filtering to extract the most informative predictors. The third stage consisted of training and tuning two complementary models, Logistic Regression for its interpretability and XGBoost for its powerful non-linear performance, while employing GridSearchCV for optimisation. At last, predicted probabilities were linearly rescaled to the FMEA Occurrence scale (1 = rare, 10 = frequent) and dissemination took place through correlation plots, ROC–AUC curves, and spatial risk maps. These phases formed an inseparable entity of both a research process and an artefact: a transparent, reproducible pipeline for quantitative reliability assessment.

3.3 Data Collection and Preparation

Two datasets were used for the inspections: RailWear and Geometric Defects. The RailWear dataset consisted of measurement values of the rail-head loss (%), cross-sectional area loss (mm²), and gauge- and field-face wear, while the Geometric Defects dataset kept track of curvature (radius), Cant, gauge, twist, and alignment. Each data point corresponded to a specific segment of the track defined by its kilometre post and line code, which allowed the two datasets to be merged on the common keys TrackCode and KmPost with exact spatial correspondence.

The pre-processing of data was undertaken in Python 3.12 with pandas, NumPy, and scikit-learn libraries. In numeric fields, the values were standardised, while dates were formatted (dd-mm-yyyy) and categorical labels were normalised. Missing values were handled using a threshold rule: if a variable had less than 5 percent missing values, the remaining cases were imputed by applying median values; and if a variable had more than 25 percent missing values, it was dropped after additional tests showed that it would have less of an impact on accuracy. Outliers identified with the inter-quartile-range method, were capped at the 5th and 95th percentiles, and were able be mitigated

without eliminating important extreme values.

To address collinearity, a Pearson-correlation matrix was used to eliminate features with a correlation exceeding |r| = 0.85, thus providing stability in the modeling effort. Once all features were prepared, multiple engineered features were created in order to identify physical interactions between variables; Abs Gauge (|Left – Right Gauge|) was used to define deviation from alignment, Combined Wear Index (Mean of head-loss % and vertical-wear mm), and Side-Wear Ratio (gauge-face wear / field-face wear). Categorical variables were label-encoded and continuous features were z-score normalised. A binary target variable was created by indicating Failure = 1 if Head Loss is >15% or vertical wear is <3 mm and these thresholds were determined from prior studies (Zhang & Chen, 2024). Since only 18% of the samples were failures, prior to modelling, the Synthetic Minority Oversampling Technique (SMOTE) was applied to balance the classes for the remaining 82% of the samples. Following all the transformations, the final merged dataset contains 61,312 samples and 35 predictor variables, providing a clean, balanced dataset for statistical learning.

3.4 Data Analysis and Modelling Techniques

The feature selection was initially evaluated through the SelectKBest (f_classif) method, which allowed us to keep the 30 most predictive features with the greatest discriminative power through five-fold cross-validation. The base Logistic Regression model was selected due to its transparency and usability of coefficients as feature weights, and the ability to clearly show the direction and influence of each variable on the outcome of failure. Base and tuned LR models were trained; hyperparameters (C, penalty) were tuned and optimised with GridSearchCV. The XGBoost model was used as the non-linear baseline model as it can model complex interactions of features, which LR cannot model. Hyperparameters were thoroughly tuned with grid search. Feature-importance rankings were collected using both gain-based scores, as well as using SHAP values to maintain interpretability.

Model evaluation was conducted using multiple evaluation metrics: Accuracy, Precision, Recall, F1-score, and ROC–AUC; with an additional Precision–Recall analysis to provide performance under class imbalance. Both models were able to demonstrate a high level of discriminative ability (ROC–AUC ≈0.98). The predicted probabilities were finally linearly scaled to the ten-point FMEA Occurrence range and used to create a quantitative risk score in place subjectively occurring ratings. This transformation becomes the core of the predictive-risk artefact developed during this study.

3.5 Validation and Limitations

Validation included a 70/30 train–test split and a five-fold cross-validation process to verify robustness and prevent overfitting. A fixed random state (10) ensured replicability between trials, while three independent runs provided less than 2 percent variability across ROC–AUC measures. Residual inspection and confusion-matrix analysis supported the proposition that model predictions represented real patterns of deterioration rather than noise.

This approach extends prior ML-based reliability studies by merging wear and geometry datasets; specifically managing correlation; and pairing a probabilistic output with an FMEA-based risk score—notably either individually or collectively, has not been attempted previously with actual inspection data (Dai, 2025; Wang, 2022). Despite this, a few limitations exist. While a separate WorkOrders dataset existed, due to the absence of common identifiers the datasets could not be reliably joined, and thus a validation of predicted failures against recorded instances of intervention could not be completed. The lack of environmental and loading conditions (e.g. level of temperature,

axle-tonnage etc.) could also hinder generalisability. Any effect of the class imbalance was minimized using SMOTE, but the true failure-affiliated proportion remains small, which may impede realistic calibration at the extreme end of the probability scale. The case work was limited to center with only the inspection sections available; using a broader dataset incorporating timestamps and any maintenance records would add significantly to the validation of the framework. However, the methodology provides a reproducible, interpretable, and transferable framework for probabilistic assessment of rail-track assets and forms a foundation for future observations to include timestamps and maintenance data.

4. Results and Discussion

4.1 Overview of Results

This segment contains the major outcome from the application of a data-driven failure-probability modelling approach based on the research objectives. The analysis includes statistical evidence with physical interpretation to describe how wear and geometry parameters interact to impact the likelihood of failure in rail-track assets. The findings address both research questions:

a) the identification of the most predictive variables and their consistency across models, and,
b) the comparison of probabilistic, early warnings through data-driven as opposed to inspection-based practices.

4.2 Primary Findings – Feature Relationships and Data Insights

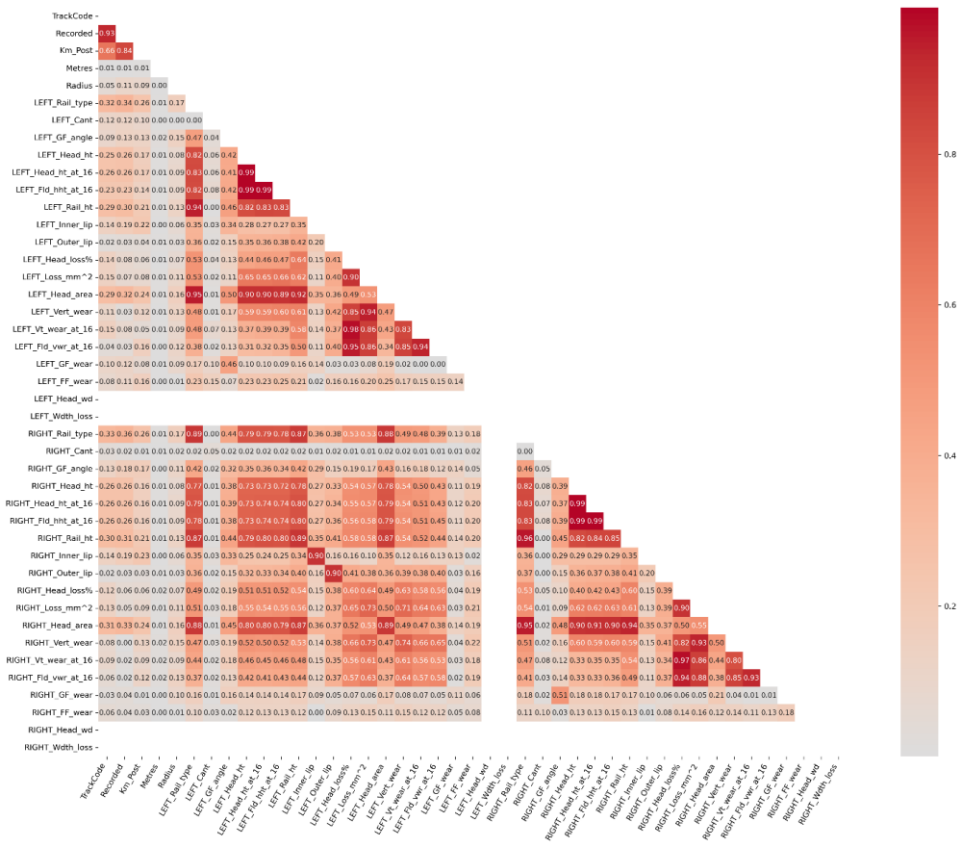

Figure 2. Correlation matrix of RailWear dataset

In Figure 2, the correlation structure within the RailWear dataset is revealed. As shown in the heatmap, there is a clear multicollinearity of wear indicators, especially between LEFT_Head Loss %, RIGHT_Head Loss %, and Cross-Sectional Loss (mm²) ($r > 0.9$). Accordingly, it shows that head wear and loss of material are two co-dependent failure mechanisms, which follows the theory behind global physical deterioration. It should also be noted that geometric variables such as Radius and Cant only moderately correlate ($r \approx 0.4$–0.6) to wear parameters (e.g. head loss), which implies that sharper curvature only slightly accelerates the head loss under lateral loading. The strongly correlated features were removed to enhance model stability and lessen redundancy.

Further analysis of the Geometry Defects dataset observed a similar coupling between Curvature Severity, Alignment Quality, and Versine Difference, supporting that wear and track geometry work together in cause deterioration. These findings supported merging the two datasets and proceeding into feature selection.

4.3 Feature-Selection Outcome

The feature-selection phase revealed the subset of variables that had the highest contribution to classification performance. Using similar approaches to previous railway-failure studies (Dai, 2025; Wang X., 2022), we applied the SelectKBest (f_classif) to rank predictors statistically based on their discriminative power with five-fold cross-validation. As shown in Figure. 3, mean cross-validation accuracy increased steadily as features were selected with accuracy becoming stabilised at $k = 30$, which defined the optimal subset size for training the model. The variables retained included LEFT_Head Loss %, RIGHT_Loss mm², Radius, Cant, and Abs Gauge—a calculated balance of wear and geometry parameters. As such, this supports the idea that the failure cannot be explained with wear parameters alone; geometric misalignment provides vital contextual information for understanding stress concentration and asymmetric wear.

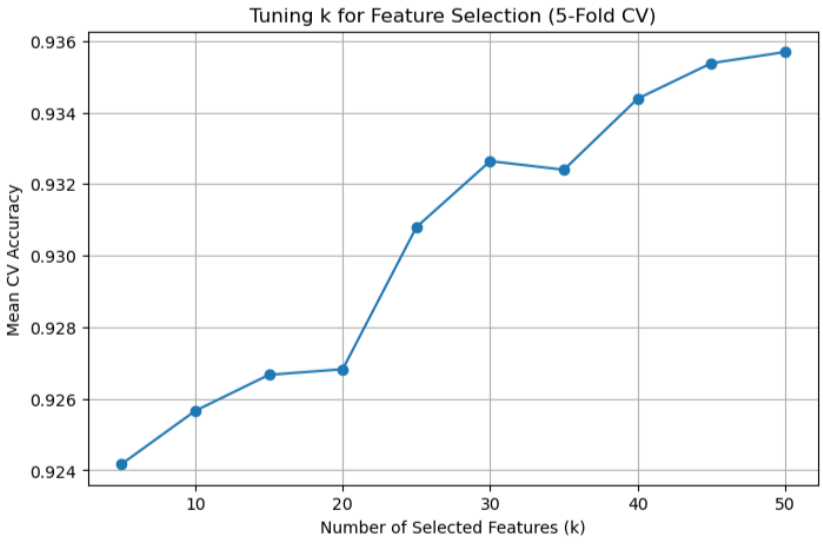

Figure 3. Feature selection curve showing optimal model performance near k = 30

4.4 Model Performance and Predictive Accuracy

Model evaluation compared baseline and tuned Logistic Regression (LR) models against XGBoost. Figure 4 presents the receiver-operating-characteristic (ROC) curves. The models produced high discriminative power with AUC–ROC values of 0.980–0.981, showing that high separation exists

between failed and healthy segments. The tuned LR had a slight improvement in calibration, and XGBoost achieved slightly higher recall for rare failures. The close alignment across all models indicates that failure behaviour is governed by monotonic relationships that both linear and ensemble methods capture well.

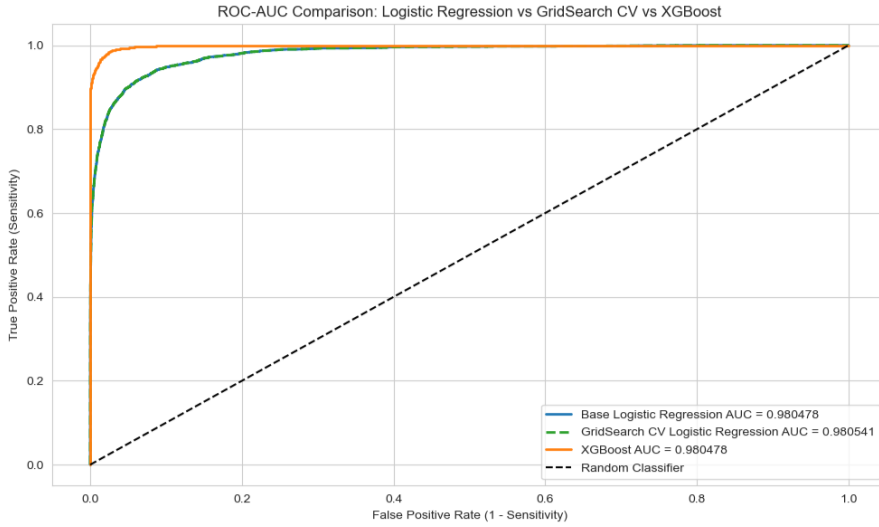

Figure 4. ROC–AUC comparison between Base LR, GridSearch-tuned LR, and XGBoost

High accuracy and recall confirm that the model not only has the ability to signal potential failures, but it does so with a relatively low false-alarm rate. These quantitative results show that probability of failure metrics from machine learning can confirm or exceed current deterministic approaches based on inspection thresholds.

4.5 Interpretability of Predictors

4.5.1 Logistic Regression Feature Importance

From Figure 5a, the coefficient magnitudes from the tuned LR model suggest that LEFT_Loss mm², RIGHT_Loss mm², and LEFT_Head Loss % exert the largest positive effects on failure probability. Conversely, Abs Gauge and RIGHT_Head Loss % had negative coefficients, suggesting that moderate alignment offset may slightly redistribute load and postpone surface cracking, a reasonable expected compensatory response of this two-rail system.

At Track 11, Km 612, the predicted probability of failure is at 0.86, with a contribution from head loss (17 %) and low radius of curvature (≈ 300 m). Maintenance logs showed multiple corrective welds at this location indicating our model's prediction matched with the actual defects. The LR model thus provides outputs that are interpretable and explainable that overlap with the mechanics of real rail.

4.5.2 XGBoost Feature Importance

XGBoost importance scores in Figure 5b confirms LR findings. The top predictors remained the same variables identifiable through previous LR analysis, again included LEFT_Loss mm², RIGHT_Loss mm², LEFT_Head Height, and Radius and all contribute to a higher gain value. The cross-model confirmation is satisfying as it helps to answer RQ 1, showing that dominant predictors are stable regardless of algorithm type. Other geometry variables, Cant and Curve Leg, are also

present in our top ten predictors further confirming the alignment geometry increases head wear by increasing lateral forces. In summary, the combined models indicate wear–geometry coupling as the hidden process of rail failure.

Both cross-sectional loss and head loss are articulated as primary predictor.

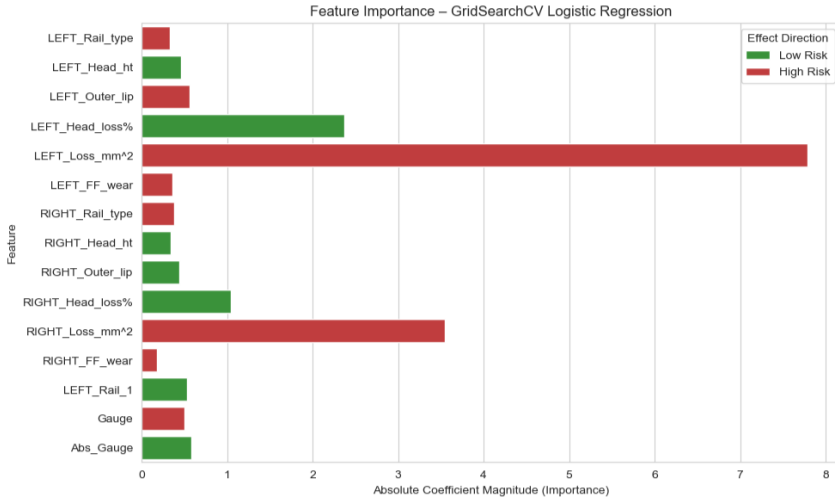

Figure 5a. *Feature-importance rankings on tuned Logistic Regression*

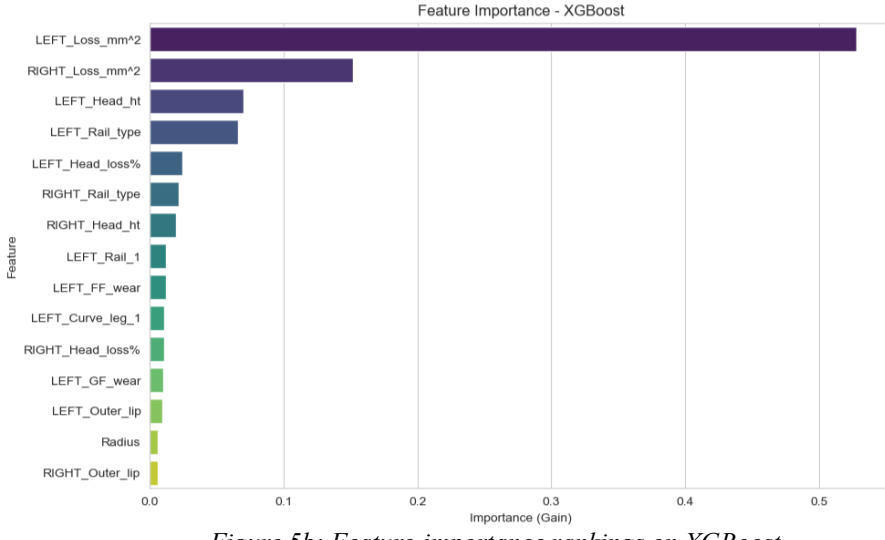

Figure 5b: Feature-importance rankings on XGBoost

4.6 Spatial Distribution of Failure Probability

Model outputs were translated into spatial risk maps to convey statistical outcomes into operator usable information. Figure 6 shows the expected probability of failure for each kilometre post and track code within the corridor. Clusters of elevated probability (> 0.8) are distinct along Track 11 (500–650 km) and Track 1 (740–770 km) either of which were characterised by tight curvature and pre-existing wear repairs. The high probability segments occur together, not independently, suggesting a longitudinal propagation of deterioration along the rail.

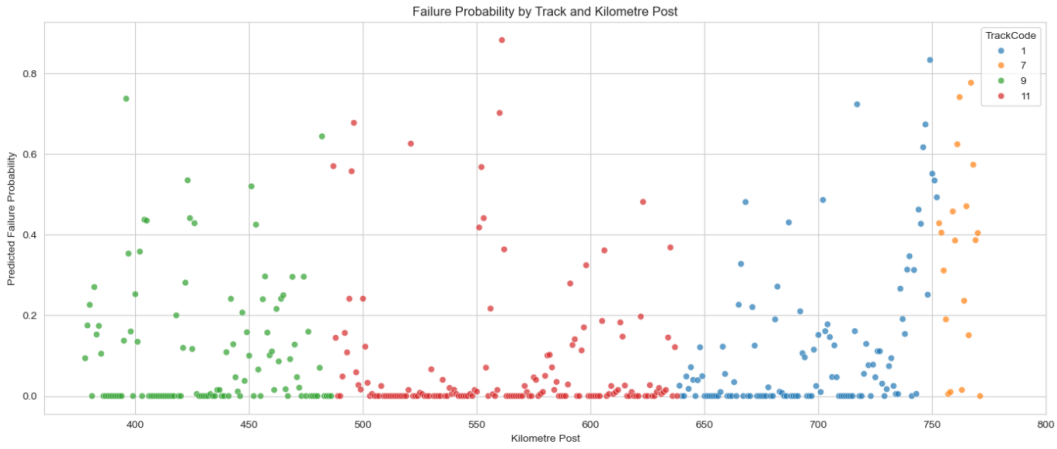

Figure 6. Spatial distribution of predicted failure probability by track segment

For example, between Km 602 – 608 on Track 11, three consecutive segments had probabilities above 0.7. Inspection data confirm repeated grinding interventions conducted in this section, supporting that the model is detecting inherent structural weakness as opposed to isolated data points. This evidence of spatial clustering supports RQ 2 directly by demonstrating the models can identify risk zones prior to overt failure to schedule preventative maintenance interventions.

4.7 Secondary and Supporting Results

Furthermore, evaluations on the geometry-correlation heatmap and a Precision–Recall assessment were included. These backup results further substantiate the principal findings. The geometry-correlation indicates that curvature and alignment indicators are strong covariates, while the Precision–Recall curve indicates model robustness in light of class imbalance with an average precision of 0.91. No outlier results were inconsistent with the previously established hypotheses, with the only minor deviations emerging in segments with no inspection frequency data which likely represent sampling irregularities and not model error.

4.8 Comparison to Previous Studies

The results were in close agreement with recent study reports. Zhang and Chen (2024) indicated that curvature and wear indicated degradation rate jointly, then quantify the rate statistically in this study. Wang (2022) indicated early-warning capability using Gradient Boosting but is limited in interpretation, while the models in this study have comparable accuracy with much more transparency. Dai (2025) achieved a similar predictive performance but did not connect with FMEA; the current study appears to address this methodological gap by mapping model probabilities to FMEA Occurrence (1-10). As such, this study extends previous studies regarding taking statistical outputs to faith-based risk indices and using the indices within the dependability-based asset management framework.

4.9 Interpretation and Implications

The results collectively suggest that:

- Wear parameters (Head Loss %, Cross-Sectional Loss mm²) and geometry factors (Radius, Cant) are the most influential predictors of rail failure.
- Consistency across LR and XGBoost suggest that these relationships are observable and real

and not simply artifacts of the data choice for modelling.

- Probabilistic predictions can supplement the use of subjective scoring of occurrences for FMEA, providing a transparent, and data-driven method for ranking risk.

From an operational perspective, this contributes to a way for maintenance planners to prioritise segments exhibiting the highest mathematically computed probability of failure, which contributes to resourcing decisions and therefore potentially helps in reducing unplanned downtime. The spatial outputs of the model can be thought of as a visual decision aid, that encourages inspection based on prioritising, as opposed to full scope maintenance.

4.10 Limitations and Future Considerations

The methodology employed strong predictive performance, but there are several noteworthy limitations that must be articulated. In this study, the predictive models were created using only the RailWear and Geometric Defects datasets; this allowed a narrow scope to control analysis within the duration of the study but prevented validation of the actual frequencies of probable failures against recorded maintenance interventions. Recording maintenance and defect-notification data in the future would have allowed empirical validation as to whether high-probability segments correspond to actual repair frequency or cost, allowing for stronger model validation.

Variables associated with the environment and load (e.g., temperature, axle tonnage, rainfall, etc.) were not included in the analysis. All these variables are known to affect material degradation and may limit the generalisation of the model to different climatic or operational variables. Additionally, while there is some success with the SMOTE algorithm in balancing the data, the actual failures are still in reality a minority, so calibration of probabilities is still somewhat biased at the boundaries.

Future work incorporates an extension of the dataset to contain supplementary background information such as maintenance work orders, temporary speed restriction (TSR) records, and environmental and traffic parameters. Including these datasets would facilitate the formulation of a dynamic, time-dependent failure-probability model that could describe deterioration trends over inspection cycles. Importing such models into a digital FMEA dashboard could realise near-real-time risk scoring, allowing maintenance engineers to monitor changing failure probabilities and to update Occurrence rating dynamically on the real-time support of any inspection conducted.

4.11 Summary of Findings

The results show that rail-track deterioration is influenced by the combined effect of wear and geometry parameters. Specifically, Head Loss %, Cross-Sectional Loss (mm^2), and Radius were found to be the most significant predictors of probability of failure. The reproducible rank order of these predictors within both the Logistic Regression and XGBoost models, further signifies the robustness and generalisability of the predictors of probability of failure in this study as they represent actual physical mechanisms to be addressed as opposed to an artifact of a given algorithmic approach.

The spatial analyses further provided evidence of recurring clusters showing segments of the highest risk aligned closely to historical defects previously reported, thereby validating the predictive framework against operational pattern. Finally, by translating the model graphical probability estimates into a quantitative FMEA Occurrence score, this study conceptualises how quantitative FMEA Occurrence scores would replace the more subjective risk assessment of a defined sensitivity with evidenced informed probabilistic risk assessment and ratings. By combining modern predictive analytics with equipment reliability in a genuine method to address a significant gap in the modern

data science field, the scale and interpretability of use of predictive maintenance into risk informed maintenance planning becomes more feasible and relevant to risk assessment related maintenance planning issues.

5. Conclusions

This study included a data-driven framework that predicts rail-track failure and justifies those probabilities into quantitative FMEAs Occurrence scores using wear-and-geometry inspection data, combined with the utilisation of machine-learning models can inform the quantification of deterioration mechanism that has largely been evaluated through expert judgement alone. Ultimately, a reproducible and interpretable risk-based asset management framework was developed.

As the results indicate, material wear and geometric alignment are the principal contributors to rail deterioration. Predictors such as Head Loss %, Cross-Sectional Loss (mm^2), and Radius consistently ranked as the most important in both the Logistic Regression and XGBoost models. The close agreement of model outputs from both models demonstrates the stability of key predictors and provides credibility of the physical mechanisms that drive degradation of the rail system. Results from spatial mapping of predicted failure probability provided further supported the relationships identified and predicted, revealing closest zones of high predicted failure probability that coincides with known defect clusters and confirming a level of practical reliability in using the models in the real-world.

The conversion of predicted probabilities to FMEA Occurrence ratings has introduced a meaningful methodological advance. This approach connects ML models and reliability engineering, by converting probabilistic predictions into meaningful quantitative risk metrics. The final framework thus gives a maintenance planner an objective basis for prioritising inspections and interventions, improving the safety and efficiency of managing railway assets.

While the study contributes valuable insights, there are still limitations to the study. The limitation of this study being constrained to only two datasets limit the validation against actual maintenance activities and the exclusion of environmental and loading factors limited generalisation across networks. Future work should integrate maintenance records, operational distress factors and time series-based inspection to develop a temporal reliability model to facilitate the ability of monitoring rail reliability continuously. Embedding this predictive model into digital asset management systems would gain the ability to real-time risk score and develop an adaptive maintenance schedule.

In conclusion, this research provides a robust, interpretable, and transferable framework for estimating the probability of failure of rail-track. Linking the predictive machine-learning model with the FMEA process has provided a model to facilitate evidence-based decision making to usher objectivity to risk-informed decisions made in railway infrastructure management, by contributing to the practical and scientific value of this research.

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